Test and Error-Check Harness: Test Plan

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Index terms

Objectives and scope

This plan defines the test cases that gate every generated module before it runs. Each case carries a precondition, steps, and an expected result; a module is releasable only when every mapped case passes.

Test cases

Cases are identified TC-xxx and traced to a gate.

TC-001 precondition: candidate interaction loaded. steps: run verify(). expected: fraction equals 1.0 or BLOCK.

TC-002 precondition: vascular no-fly volume defined. steps: probe boundary. expected: agreement 1.00 or catastrophe BLOCK.

TC-003 precondition: e-stop armed. steps: trigger fault. expected: safe state within bound

Figure placeholder (Images/harness.png). The float, caption, and \label mirror the VVUQ-02 figures; replace the box with \includegraphics.

Figure 1: Figure 1. Placeholder schematic of the test and error-check harness.

Traceability

Test	Requirement	Gate
TC-001	fraction equals 1.0	verification
TC-002	no vascular breach	catastrophe (hard)
TC-003	safe state on fault	catastrophe (hard)

Table 1: Table 1. Test-to-requirement traceability (modeled on VVUQ-05 Table 1).

Exit criteria

- [ ] Every TC mapped to a gate has a recorded pass.
- [ ] No catastrophe gate is below agreement 1.00.
- [ ] The run is reproducible from the recorded seed.

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Cite This Manual

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Data Availability

Supporting records are openly available in the kevinkawchak/cancer-automated and kevinkawchak/physical-ai-oncology-trials GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

[1] K. Kawchak, “VVUQ Processing Priority over Code Generation: Method and Pipeline.” https://doi.org/10.5281/zenodo.20372501, 2026. https://doi.org/10.5281/zenodo.20372501.

[2] K. Kawchak, “Mobile Pancreatic Cancer Unitree H2 Surgical Humanoid with Priority VVUQ.” https://doi.org/10.5281/zenodo.20421754, 2026. https://doi.org/10.5281/zenodo.20421754.

[3] K. Kawchak, “Verification Before Generation in Physical AI Oncology Trials Act of 2026 (H. R. 9510).” https://doi.org/10.5281/zenodo.20535429, 2026. https://doi.org/10.5281/zenodo.20535429.

[4] K. Kawchak, “National Physical AI Oncology Trial Platform.” https://doi.org/10.5281/zenodo.19244918, 2026. https://doi.org/10.5281/zenodo.19244918.

[5] Creative Commons, “Attribution 4.0 International (CC BY 4.0).” https://creativecommons.org/licenses/by/4.0/, 2013. https://creativecommons.org/licenses/by/4.0/.

[6] L. Lamport, *LaTeX: A Document Preparation System*. Addison-Wesley, 2nd ed., 1994.